

Work Experience - Roles and Responsibilities

AVANTEL, Visakhapatnam/Hyderabad, India

Embedded Engineer - Design & Development Department

August 2023 – Present

Technical Skills:

Device Architecture, C Program, Python, Bare metal, Therapy Algorithms development, ISO 80601-2-70, IEC 62304, Pressure/Flow/Volume Control Algorithms, Pattern Analysis, Prediction and Prevention, UART, I2C, SPI, QSPI, SDHC, Sensor Calibration, Sensor Filtering and Measuring Techniques, Timing Sequence Execution and validation, Interrupt driven program, Boot loader, Flash Memory, OTA. Run Time Self Programming, Network Packet Analysis, Wi-Fi Protocol Debugging, Ubuntu - Sniffer, Wire Shark, Multidisciplinary Skills, Coordination with UI/UX , Software team - ICD, Sales and Production team.

Key Responsibilities & Achievements:

- **Designed Architecture** for medical devices of CPAP, BiPAP and HFNC devices.
- **Firmware/Embedded Team Lead** for CPAP, BIPAP, and HFNC medical device development.
- Led **end-to-end product development** from prototype to production in compliance with **ISO 80601-2-70** and **IEC 62304**.
- Designed and implemented **therapy control and safety algorithms** to improve patient comfort, adaptability, and device performance.
- Developed and validated therapy modes: **CPAP, Auto CPAP, APAP, BIPAP-S, BIPAP-T, BIPAP-ST, and HFNC**.
- Programmed **PIC18F, PIC24F, and PIC32** microcontrollers using **Embedded C (bare-metal)** for real-time, low-latency control.
- Implemented flow **pattern detection algorithms** of snoring, flow limitations and varies breath patterns.
- Implemented Algorithms to **predict and prevent** sleep events.
- Designed **blower motor control algorithms** using **open-loop, closed-loop, feedforward, and fuzzy logic** techniques.
- Performed **pressure and flow sensor calibration** using correlation and regression methods for clinical-grade accuracy.
- Designed to **hardware architecture**, ensuring tight firmware–hardware integration.

- Designed and implemented **GUI displays**, including display configuration and user interaction logic.
- Implemented **I2C, SPI,SDHC, and UART** communication for sensors, EEPROM, RTC, SD card, and display synchronization.
- Developed **SDHC SD card therapy data logging** with real-time timestamping using RTC.
- Enabled **Wi-Fi connectivity** for secure therapy data transfer to **mobile applications and cloud platforms**.
- Coordinated with the **mobile application team**, prepared **ICD**, and implemented **device–app–cloud communication**.
- Implemented **OTA firmware updates** via mobile app and cloud with robust **fail-safe mechanisms**.
- Implemented **Device Firmware Update processes using QSPI Flash**, supporting controller and display firmware updates.
- Implemented **state-machine–based firmware architecture** with dynamic task behavior.
- Performed advanced **firmware and hardware debugging**, including timing analysis, profiling, schematic review, and root-cause analysis.
- Designed **system-level and low-level architectures** and guided interns in **SDLC documentation** as per IEC 62304.
- Collaborated with **mechanical, PCB, hardware, and software, Store and QA, Sales team**, leveraging a strong mechatronics background.
- Excelled in **Device debugging** i.e Firmware, Hardware and Network Packet Analysis
- In the device find the root causes for TCP hand shake fail, Hardware Reset, External INT Triggers, Uart call back handlings misfires, Control System Behaviour etc.
- **Led and mentored interns and junior engineers**, ensuring technical quality and on-time delivery.
- Ensured **regulatory compliance, patient safety, and device reliability** throughout development.
- Developed a **snoring and flow restrictor simulator** capable of replicating human snoring patterns using mechatronics principles.
- Designed and built a **human breath simulator and snoring devices**, saving the company **12+ lakhs** by replacing external testing equipment and time.

HBL Power Systems Ltd, Shamirpet, Telangana, India

Senior Engineer EME - Development Division

September 2021 – April 2023

Technical Skills:

PLC Programming (IEC 61131), HMI Design, Motion Control (SERVO, STEPPER, DC, BLDC, VFDs), IoT (ESP32-S2, Arduino Pro), Embedded Systems, Pneumatics, Electrical Circuit Design, SolidWorks, 3D Printing, BOM Management, Computer Vision (Jetson Nano, Raspberry Pi), Sensor Integration, Portenta H7, STM32, Motion Control Algorithms, IMU, Algorithm developer.

Key Responsibilities & Achievements:

- Programmed **PLCs** (compliant with IEC 61131) to manage complex machine operations and sequences.
- Implemented **Ampoule Making Machine** logic from reference code, including speed-controlled DC motors, burners, and pneumatic cylinders.
- Designed intuitive **Human-Machine Interfaces (HMIs)** to enhance operator usability and safety.
- Demonstrated strong debugging skills with structured logic breakdown and **root cause analysis** for rapid issue resolution.
- Spearheaded **Industry 4.0 initiatives** across departments, integrating **IoT** and **data analytics** to enhance productivity and product quality.
- Enabled **real-time monitoring and alarm systems** via smart factory concepts.
- Designed and implemented **motion control systems** using SERVO, STEPPER, DC, BLDC motors, and **VFDs** for precise and efficient operation.
- Developed **low-cost automation solutions** that significantly reduced manual efforts and increased throughput.
- Selected and integrated **pneumatic components**, including proportional valves; developed optimized pneumatic circuits for automation.
- Demonstrated proficiency in **sensor technology**, selecting and integrating appropriate sensors for varied industrial applications.
- Designed and developed **electrical circuits** and **control systems**, ensuring robust operation and minimal downtime.
- Conducted extensive **machine testing and validation**, ensuring all equipment met performance and safety standards prior to dispatch.
- Managed project budgeting and cost estimations, ensuring financial alignment with deliverables.
- Coordinated with **suppliers and vendors** to procure high-quality components at competitive prices.
- Created comprehensive **Bills of Materials (BOMs)** for electronic and electrical procurement and assembly.
- Applied **SolidWorks** and **3D printing** for concept development, rapid prototyping, and custom mechanical component fabrication.

- Implemented code in python-**computer vision** using Raspberry Pi and Jetson Nano to detect and fill 0.6 mm holes with high accuracy.
- Designed and deployed **embedded IoT systems** enabling seamless machine-to-machine and machine-to-cloud communication.
- Developed cloud solution for RH monitoring in glove boxes, with a budget of 80K and supplier quotation is 7.5Lakhs.

VSRMKP Technologies Pvt Ltd, Hyderabad, Telangana, India

Graduate Engineering Trainee - Contract

January 2021 – June 2021

Technical Skills:

BLDC Motor Control, Li-ion Battery Management (1S to 7S BMS), Embedded C, MATLAB Modeling, Embedded Systems, Battery Charging/Discharging Algorithms, E-bike System Integration, Kalam-filter

Key Responsibilities & Achievements:

- Designed and built an **Electric Bike (E-bike)** powered by a **BLDC hub motor** and **Li-ion battery pack**, focusing on high performance, durability, and energy efficiency.
- Assembled the **BLDC hub motor**, ensuring precise mechanical alignment and secure power connections for optimized torque and motor efficiency.
- Developed and tested **embedded software** to manage the **charging and discharging cycles** of the Li-ion battery using a **1S-7S Battery Management System**, improving battery safety and lifespan.
- Simulated various E-bike functionalities using **MATLAB models**, including **speed control**, **battery state monitoring**, and **energy optimization**.
- Using ESP32 modules implement real-time control and monitoring features for vehicle operation.

Mahatma Gandhi Institute of Technology, Hyderabad, Telangana, India

Technical Assistant - Contract

November 2019 - April 2020(Sponsored by JNTUH)

Technical Skills:

Mechatronics System Design, Sensor Integration, Embedded C, Microcontroller Programming, DC Gear & Servo Motor Control, I2C & SPI Protocols, Human-Machine Interface (HMI), Precision Welding Automation

Key Responsibilities & Achievements:

- **Led the mechatronics-based design** of an **Automatic TIG Welding System**, ensuring coordinated functionality among motors, sensors, and control systems for consistent weld quality.
- Integrated various **sensors** (encoders, position sensors) and **actuators** (DC gear motors, servo motors) to enable real-time precision control of the welding head.

- Developed embedded firmware for the system's **microcontroller**, enabling real-time data processing, sensor feedback handling, and closed-loop motor control.
- Implemented **I2C and SPI communication protocols** to facilitate seamless interaction between sensors, actuators, and the main controller.
- Designed an **intuitive display interface** to allow users to monitor and control system parameters during welding operations, enhancing usability and operational efficiency.
- Maintained a strong focus on **constant linear momentum control** to achieve uniform weld quality and reduce defects during TIG welding processes.